Lance G. Montenegro

BSIT 4

A Data Flow Diagram (DFD) shows how data moves within a system using symbols and clear naming rules. It helps us understand, analyze, and design systems effectively. A DFD has four main elements: processes, data flows, data stores, and entities. Processes represent tasks where data is transformed. In a Level 0 DFD (context diagram), the process represents the system as a whole, like “Job Portal.” Level 1 diagrams break it into smaller tasks, each named with a verb, adjective, and noun, such as “Verify User Login.” A process must have at least one input and one output flow. Data flows, shown as arrows, represent the movement of information and are labeled with singular nouns. They should not split into multiple flows. Data stores hold information like files or databases and appear only in Level 1, while entities represent data sources or destinations and are always nouns.

To create an accurate DFD, certain Do’s and Don’ts must be followed. Don’ts include connecting entity to entity, data store to data store, entity to data store directly, or data store to entity directly. Do’s include connecting entity to process to entity, data store to process to data store, entity to process to data store, data store to process to entity, and process to process. Following these rules ensures the DFD clearly shows how data moves through the system and avoids confusion, making it easier for analysts, developers, and stakeholders to understand and design the system effectively.